Norton Children's Neuroscience Institute
Affiliated with UL School of Medicine

Migraine Management Pathway: Inpatient v.1.0 (2020)

Not to be a substitution for appropriate clinical judgment

Intended for acute status migrainosus not for chronic daily headaches. For chronic daily headache or chronic migraine, it is not reasonable to get headache to 0/10 and thus treatment of a chronic headache should be case-by-case. Please consult or contact the Neurology Team concerning these patients.

Confirm medication history. Recent medication may alter migraine management

Inclusion Criteria:
Patient experiencing a migraine
As defined by International
Headache Society (HIS)
Exclusion Criteria:
Secondary Headache or
No previous dx of:
Migraine with brainstem aura
Hemiplegic migraine
Retinal migraine

Confirm medication is not contraindicated for specific patient (i.e. pregnancy, HTN, etc.)

Step 1

1. **Obtain lab work: urine HCG, CBC, CMP, mag level**
2. **Give NS Bolus 20mL/kg (MAX 1000mL)**
3. **Give Benadryl 1mg/kg/dose IV (max 25mg/dose) (this must be given 15-30 minutes prior to administering Step 2 meds to help prevent dystonic reaction when giving Compazine)**

15 minutes

Step 2

2. **Give *Prochlorperazine (Compazine) 0.15 mg/kg/dose IV; MAX 10 mg/dose (may substitute Reglan if needed 0.15mg/kg/dose IV MAX 10mg/dose)**

   Ketorolac (Toradol) 0.5 mg/kg/dose IV; MAX 30 mg/dose (not to be give if patient received NSAID within the past 6 hours)
3. **Begin ½ maintenance fluids**

30-60 min

Step 3

- **Reassess pain level. If Headache resolved, may be discharged home**
- **If headache still persists and HCG test is negative give VPA bolus of 20mg/kg/dose IV over 5 minutes (MAX dose 1000mg)**

30-60 min

Step 4

- **Reassess pain level. If headache resolves may be discharged home with oral VPA 10mg/kg/day divided BID (MAX 500mg daily) x1-2 weeks with neurology f/u appointment in 2-4 weeks. Please notify neurology so f/u appointment can be made the following day.**
- **If headache still persists Consult Neurology for admission and further management**

Admission

Step 5

- **Admit to neurology**
- **Continue ½ maintenance fluid**
- **May repeat migraine cocktail x2 doses overnight Q6H for continued pain (MAX total of 3 doses since arrival to ED)**
  - **Give Benadryl 1mg/kg/dose IV (max 25mg/dose) (this must be given 15-30 minutes prior to giving Compazine and Toradol**
  - **Give *Prochlorperazine (Compazine) 0.15 mg/kg/dose IV; MAX 10 mg/dose (may substitute with reglan (0.15mg/kg/dose MAX 10mg/dose if needed) AND**
  - **Ketorolac (Toradol) 0.5 mg/kg/dose IV; MAX 30 mg/dose AND**
- **Consider behavioral therapy consult as well as acupuncture consult**

NEUROLOGY will assess and consider giving mag vs DHE as next line treatment

Step 6-7

Magnesium Sulfate: 30 mg/kg/dose IV over 30 minutes; MAX 2000 mg/dose

If Mg level is >2.3, bolus therapy should not be given since the patient may reach a toxic level during the bolus.

Magnesium may decrease blood pressure.

- Patients should have blood pressure measured before administration of IV magnesium bolus. If below normal range, reconsider therapy or administer normal saline bolus prior to magnesium and recheck blood pressure during as well as after the infusion.

2 Hours

1. Repeat Mg level
2. Additional bolus of Magnesium Sulfate: 30 mg/kg/dose IV over 30 minutes; MAX 2000 mg/dose if Mg level ≤ 2.3

- May consider repeat VPA bolus of 20mg/kg/dose IV over 5 minutes (MAX of 1000mg)

Step 6-7

Dihydroergotamine (DHE): DHE Protocol for up to 7 administrations Q8H

- Contraindications: recent use of MAOIs, antiretroviral medications, or macrolide antibiotics; triptans w/in 24h; known cardiac conditions or vasculopathy; headache not c/w migraine w or wo aura
- Administer Anti-emetic +/- diphenhydramine prior to DHE administration
- Give DHE (if less than 40kg give 0.25mg for initial dose then 0.5mg for subsequent doses; if greater than 40kg give 0.5mg for initial dose then 1mg for each subsequent dose) (MAX DOSE 10mg)
  - B/P checked Q15 minutes before dose administration and 15 minutes after completion
  - Assess for adverse effects of DHE including but not limited to chest pain, nausea, vomiting, delayed cap refill, or paresthesia
- Often, headaches will not begin to respond until the 5th dose. Some patients may complain of a worsening headache at beginning of DHE protocol. *This is not a contraindication to continued use*.
- It is recommended to give 1-2 doses past a 0/10 headache for maximum benefit

D/C

Discharge home with migraine prevention and treatment strategy and f/u appointment in Child Neurology Headache Clinic. Headache goal is 0/10 for acute status migrainosus

International Headache Society (IHS) Diagnostic Criteria

ICHD-3

Migraine without aura:	Migraine with aura:
• At least 5 attacks • Duration 4-72 hours* • At least 2 of the following: 1. Unilateral location* 2. Pulsating quality 3. Moderate to severe pain intensity 4. Aggravation by or causing avoidance of routine physical activity • At least one of the following: 1. Nausea and/or vomiting 2. Photophobia or phonophobia	• At least 2 attacks • ≥1 fully reversible aura symptoms: 1. visual; 2. Sensory; 3. Speech and/or language; 4. Motor; 5. Brainstem; 6. Retinal • At least 3 of the following 6 characteristics: 1. at least one aura symptom spreads gradually over ≥5 minutes 2. two or more aura symptoms occur in succession 3. each individual aura symptom lasts 5-60 minutes 4. at least one aura symptom is unilateral 5. at least one aura symptom is positive 6. the aura is accompanied, or followed within 60 minutes, by head

*In children and adolescents (aged under 18 years), attacks may last 2-72 hours (the evidence for untreated durations of less than two hours in children has not been substantiated).

*Migraine headache in children and adolescents (aged under 18 years) is more often bilateral than is the case in adults; unilateral pain usually emerges in late adolescence or early adult life.

Exclusionary Criteria:

- If the patient presents with a **new onset** migraine with brainstem aura, hemiplegic migraine, or retinal migraine (see below), please contact on-call child neurology. **These may be symptoms of a stroke and require urgent evaluation.** If the patient has a known/established diagnosis of one of these migraine types, pathway can be used with discretion.
- Secondary headache: Per ICHD-3 definition: "Another disorder scientifically documented to be able to cause headache has been diagnosed." Primary headaches include but are not limited to migraine and tension-type headaches.
- This is intended for acute status migrainosus **not for chronic daily headache or chronic migraine.** For chronic daily headache or chronic migraine, it is not reasonable to get headache to 0/10 and thus treatment of an acute on chronic headache should be case-by-case. Please contact neurology team for further guidance

Migraine with brainstem aura

- **Description:**
  - Migraine with aura symptoms clearly originating from the brainstem, but no motor weakness.
- **Diagnostic criteria:**

a) Attacks fulfilling criteria for 1.2 *Migraine with aura* and criterion B below

b) Aura with both of the following:

c) at least two of the following fully reversible brainstem symptoms:

d) a) dysarthria[1]

e) b) vertigo[2]

f) c) tinnitus

g) d) hypacusis[3]

h) e) diplopia[4]

i) f) ataxia not attributable to sensory deficit

j) g) decreased level of consciousness (GCS ≤13)[5]

k) no motor[6] or retinal symptoms.

Hemiplegic migraine

- **Description:**
  - Migraine with aura including motor weakness.
- **Diagnostic criteria**

a) Attacks fulfilling criteria for 1.2 *Migraine with aura* and criterion B below

b) Aura consisting of both of the following:

c) fully reversible motor weakness[1,2]

d) fully reversible visual, sensory and/or speech/language symptoms.

Retinal migraine

- **Description:**
  - Repeated attacks of monocular visual disturbance, including scintillations, scotomata or blindness, associated with migraine headache.
- **Diagnostic criteria:**

A. Attacks fulfilling criteria for 1.2 Migraine with aura and criterion B below

B. Aura characterized by both of the following:

1. fully reversible, monocular, positive and/or negative visual phenomena (eg, scintillations, scotomata or blindness) confirmed during an attack by either or both of the following:
   - – clinical visual field examination
   - – the patient's drawing of a monocular field defect (made after clear instruction)
2. at least two of the following:
   - – spreading gradually over ≥5 minutes
   - – symptoms last 5-60 minutes
   - – accompanied, or followed within 60 minutes, by headache

C. Not better accounted for by another ICHD-3 diagnosis, and other causes of amaurosis fugax have been excluded.

Contraindications and Considerations

Confirm the patient does not have contraindications to the use of medications recommended in the Inpatient Migraine Management Pathway.

1. This is intended for acute status migrainosus **not for chronic daily headache or chronic migraine.** For chronic daily headache or chronic migraine, it is not reasonable to get headache to 0/10 and thus treatment of an acute on chronic headache should be case-by-case. Please contact neurology team for further guidance
2. If the patient presents with a new onset migraine with brainstem aura, hemiplegic migraine, or retinal migraine (see below), please contact on-call child neurology**. These may be symptoms of a stroke and require urgent evaluation.** If the patient has a known/established diagnosis of one of these migraine types, pathway can be used with discretion.
3. Avoid using vasoconstricting agents (like triptans and DHE) if uncontrolled HTN, cardiovascular, cerebrovascular, or peripheral vascular disease.
4. Also avoid using vasoconstricting agents if migraine with brainstem aura, hemiplegic migraine, or retinal migraine.
5. Pregnancy or breastfeeding
6. If considering NSAIDs, last dose of NSAID within 6 hours or if history of GI bleeding
7. If considering triptans, last dose of a triptan within 2 hours or more than 2 doses of a triptan within a 24 hour period
8. If considering DHE, last dose of triptan within 24 hours prior DHE
9. If hypotensive, consider IV fluid bolus prior to administration of medications like dopamine receptor antagonists
10. Severe hepatic and renal disease
11. Secondary headache diagnosis (e.g., headache attributed to idiopathic intracranial hypertension)

* Obtain history of all medications, including recent medications taken for headache.

* Consider medication overuse with acute therapies. Overuse defined as ≥15 days per month for simple analgesics or ≥ 10 days per month for analgesics with caffeine, triptans, and opioids

Who Should Receive Treatment?

The overall mean prevalence of pediatric migraine is 9.1% and increases with age (Wober-Bengal, 2003). Patients may present with acute migraine attacks or “refractory” migraine (failing both preventative and abortive treatments). If the duration of the migraine attack exceeds 72 hours, it is defined as status migrainosus in the International Classification of Headache Disorders 3rd edition (ICHD-3). Approximately 6-7% of patients treated for headache in the emergency department or an infusion center will fail treatment and require inpatient hospitalization (Kabbouche, 2015).

Migraine Pathway

Environmental Considerations

Place patient in quiet, dim light room if possible. Prohibit use of electronic devices. Encourage patient to get up and move often and have normal sleep and eating routine.

Normal Saline Bolus

Every patient complaining of a migraine should first receive (unless contraindicated):
20 mL/kg Normal Saline (NS) bolus (MAX 1000 mL)

1st Line Medications

1st Line Medications

- Diphenhydramine (Benadryl) 1 mg/kg/dose IV; MAX 25 mg/dose.
  - Administered 15 – 30 minutes prior to prochlorperazine to decrease risk of dystonic reaction
- Prochlorperazine (Compazine) 0.15 mg/kg/dose; MAX 10 mg/dose
  - Ask if patient has had dystonic reaction before
- Administer ketorolac (Toradol) 0.5 mg/kg/dose IV; MAX 30 mg/dose immediately following prochlorperazine if patient has not had an NSAID within the past 6 hours

*Metoclopramide (Reglan) May be used in lieu of prochlorperazine if prochlorperazine unavailable (studies show prochlorperazine to be more effective than metoclopramide)

- Dosing: 0.15 mg/kg/dose IV; MAX 10 mg/dose

2nd Line Medications

Wait 30 min-1 hour and reassess pain level.
Obtain Urine HCG (if pt. > 9 yo or post menarche)
If no contraindications (liver dysfunction, pregnancy, etc.), give
VPA bolus 20mg/kg/dose IV over 5 minutes
MAX: 1000mg/dose

- Repeat “migraine cocktail” with diphenhydramine, prochlorperazine, and ketorolac (MAX 3 per 24-hour period)

3rd Line Medications/INPATIENT MANAGEMENT

- Wait 30 min-1 hour and reassess pain level. If patient still has migraine refractory to treatment, call neurology. Patient may be admitted to inpatient service. Our goal is a 0/10 headache for acute status migrainosus.
- If admitted, ensure patient is in a dim, quiet room (discuss with family and RN about prohibiting access to screen time, including electronic devices like cell phones, computers, and television). Encourage to not sleep all day but keep a consistent sleep and eating schedule and get up and move as much as possible.
- Consider following options:
  - Maintenance IV fluids
  - Magnesium Sulfate 30 mg/kg/dose to be administered IV over 30 min; MAX 2000 mg/dose
    - Obtain magnesium level prior to administration. DO NOT Administer magnesium if patient's Mg level is >2.3
    - Blood Pressure monitored prior to, during, and after infusion. Give NS bolus if hypotensive.
    - May repeat in 2 hours if repeat Mg level is ≤ 2.3, following same instructions as above
  - Repeat Depakote bolus 20 mg/kg/dose IV over 5 minutes, MAX: 1000 mg/dose
  - Dihydroergotamine (DHE): DHE Protocol for up to 7 administrations Q8H
    - Contraindications: recent use of MAOIs, antiretroviral medications, or macrolide antibiotics; triptans w/in 24h; known cardiac conditions or vasculopathy; headache not c/w migraine w or wo aura
    - Administer Anti-emetic +/- diphenhydramine prior to DHE administration
    - Give DHE 0.1mg IV once followed by 30 minutes later by second dose if tolerated
      - B/P checked Q15 minutes before dose administration and 15 minutes after completion
      - Assess for adverse effects of DHE including but not limited to chest pain, nausea, vomiting, delayed cap refill, or paresthesia
      - Continue dosing Q8H, increasing each dose by 0.1mg, with max dose of 1mg and a targeted total of 9 doses
      - Often, headaches will not begin to respond until the 5th dose. Some patients may complain of a worsening headache at beginning of DHE protocol. *This is not a contraindication to continued use*. Recommend giving once dose past a 0/10 headache for maximum benefit.
  - Consider triptan if has not been used, although less likely to be effective as more time elapses
  - Corticosteroids (insufficient data to recommend in acute treatment of migraine in the ED) but may consider:
    - Dexamethasone 0.25-0.5 mg/kg/dose
    - Methylprednisolone 1 mg/kg/dose
    - Warn patients about low but possible risk of **avascular necrosis**
  - Avoid opioids and barbiturate-containing medications
  - Consult acupuncture (can be contacted at any time during the pathway)
  - Consider early consultation of behavioral health

Discharge Criteria

- This is a highly individualized approach, and our goal is to reach headache resolution of 0/10. This is intended for acute status migrainosus **not for chronic daily headache or chronic migraine.** For chronic daily headache or chronic migraine, it is not reasonable to get headache to 0/10 and thus treatment of an acute on chronic headache should be case-by-case
- The patient should be discharged home with a migraine prevention and treatment strategy
- If patient had a good response to VPA consider discharge home with 2 week course of oral VPA 10mg/kg/day divided BID (MAX dose 500mg daily)

Migraine Prevention Strategy

- Triggers
  - Identify and avoid migraine triggers
- Fluid
  - Drink at least 6 cups (1.5L) of water daily
  - Remover or reduce caffeine from the diet
- Sleep
  - Maintain a regular sleep pattern every day
- Diet and exercise
  - Eat 3 meals a day, do NOT skip meals
  - Eat a protein source for breakfast
  - Maintain a healthy body weight
  - Exercise for 30 minutes daily
- Mind and body
  - Stress and anxiety can worsen migraines
  - Migraines can cause physical and mental stress
  - Migraines can cause a change in mood or behavior
  - Biobehavior techniques are beneficial
    - Relaxation training
    - Biofeedback
    - Cognitive behavior therapy
    - Stress management
- Supplements
  - Magnesium Sulfate 9 mg/kg/day BID (MAX 600 mg/day)
  - Coenzyme Q10 1-3 mg/kg/day daily (MAX 200 mg/day)
  - Vitamin B2 200-400 mg/day daily
  - Melatonin

Prophylaxis Criteria

1. If the patient has two (2) or more headaches per week that are associated with disability or
2. > 3 incapacitating headaches per month or
3. If the headache is predictable (ex. premenstrual migraine)

Prophylaxis – Starting doses

- **2-5 yrs**
  - Cyproheptadine 0.25 mg/kg/day QHS or divided BID (4-12mg daily)
    - Neurology typically doses QHS due to sedation SE
- **≥6 yrs**
  - Propranolol 0.5-1 mg/kg/day divided BID (Initial MAX 80 mg/day)
  - Atenolol 0.5-1 mg/kg/day divided BID (Initial MAX 50 mg/day)
  - Amitriptyline 0.25 mg/kg QHS (Initial MAX 10 mg/day)
  - Imipramine 1 mg/kg QHS (Initial MAX 10 mg/day)
  - Verapamil 2-3 mg/kg/day divided BID (Initial MAX 240 mg/day)
  - Valproic acid 10-30 mg/kg/day divided BID (Initial MAX 1000 mg/day)
  - Topamax 2-3 mg/kg QHS (Initial MAX 150 mg/day)
  - Gabapentin up to 2400mg/d divided BID-TID
  - Zonisamide 200-400mg/day
  - Levetiracetam 500-3000mg/day divided BID

Migraine Treatment Strategy

- See below for triptan dosing
- In studies Naproxen, then Ibuprofen, then tyelnol is the tier recommended for most effective in treatment of acute headache
    1. 5mg/kg Naproxen PO; MAX 500 mg/dose (Naproxen not to be used if NSAID given within past 6 hours)
    2. Ibuprofen: 10 mg/kg/dose (MAX 600 mg/dose) Q6H
    3. Acetaminophen: 15 mg/kg/dose (MAX 1000 mg/dose) Q4H

Triptan dosing and Use of Triptans in the Acute Setting

- Refer above for uses and contraindications to triptans. Patients may have taken triptans prior to arrival to the ED. IV medical therapy is likely to be more efficacious in patients with status migrainosus as triptans tend to be more helpful when given early in the course of a migraine.
- Only almotriptan (≥12 yo), rizatriptan (6-16 yo), sumatriptan/naproxen (≥ 12 yo), and zolmitriptan NS (≥ 12 yo) are FDA approved for use in children, but other triptans have been used off-label in this population.
- Consider using alternate delivery routes aside from oral medications if patients have associated nausea/vomiting
- Common side effects of triptans include chest, neck, throat pain/pressure/tightness, flushing, dizziness, and drowsiness.
- According to most recent guideline from the American Academy of Neurology and American Headache Society (Oshoui et al. Practice guideline update summary: Acute treatment of migraine in children and adolescents, 2019), the following recommendations were made regarding selection of triptans in adolescents:
    - sumatriptan/naproxen oral tablet (10/60, 30/180, 85/500 mg)
    - zolmitriptan nasal spray (5 mg),
    - sumatriptan nasal spray (20 mg)
    - rizatriptan oral dissolving tablet (5 or 10 mg)
    - almotriptan oral tablet (6.25 or 12.5 mg)

    *adjust dosing as appropriate for patient's age and weight

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